

1. colouring theorem

ref. ch1

s'pose N : poisson w/ mean measure M .

M : conditional distribution given N of form $B(N, p)$

$\Rightarrow M$ & $N-M$ are independent poisson processes w/ mean μp & $M(1-p)$

colouring with two colours

let π : poisson process on general state space S w/ mean measure M .

$\hookrightarrow$ colour the points red or green (independent)

For any $A \subseteq S$, # of points of π in A , red, green $\rightarrow N(A), N_r(A), N_g(A)$

$(N(A) \rightarrow P(M(A)))$

$N_r(A) \rightarrow$ conditional distribution given $N(A)$: $B(N(A), p)$

Then, $N_r(A), N_g(A)$ are independent poisson processes w/ distribution $P(pM(A))$ and $P(qM(A))$

And if $A_1, A_2, \dots, A_n$ are disjoint sets in S

$(N(A_i), N_r(A_i), N_g(A_i))$ $i=1, 2, \dots, n$ are independent.

$\underbrace{N_r(A_i), N_g(A_i)}_{2n \text{ random variables}}$ $i=1, 2, \dots, n$ are also independent.

$\therefore$ set of red & green points form independent poisson process w/ mean measure pM & qM

colouring theorem

let π : poisson process on S with mean measure M .

let the points of π be randomly coloured w/ K colours where probability of receiving i th colour is p_i

$\hookrightarrow$ each point's colour is independent of each other.

let π_i be set of points w/ i th colour.

then π_i are independent poisson process w/ mean measure $M_i = p_i M$.

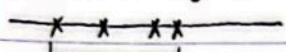
* via superposition theorem

we may also conclude that since π is superposition of π_i , $M = \sum_{i=1}^K M_i = \sum_{i=1}^K p_i M$

$\hookrightarrow$ proof by induction.

example

cars moving on a stretched road


 road = $\mathbb{R}$
points = cars $\rightarrow$ moves at different speedat some time $t \rightarrow$ snapshot of road $\rightarrow$ cars form a poisson process

(f) speed of cars are random variables, independent of one another & road, does not change w/ time, overtaking is unrestricted, takes finite # of values,

(then) let $V = \{v_1, v_2, \dots, v_K\}$ $v_1 < v_2 < \dots < v_K$ be all finite possible speeds

let p_i be the probability that the car moves at speed v_i

via colouring theorem, ... are independent

@ $t=0$: π_i : set of cars of speed v_i at $t=0$ is independent poisson process w/ mean measure $p_i \lambda(x) > 0$

via mapping theorem,

@ $t=?$: $\pi_i + v_i t$: position of cars form a poisson process $\pi_i(t) = \pi_i + v_i t$ is an independent poisson process for $i=1, 2, \dots, K$

via superposition theorem,

$\pi(t) = \bigcup_{i=1}^K \pi_i(t) = \bigcup_{i=1}^K (\pi_i + v_i t)$ is also a poisson process w/ mean measure of $p_i \lambda(x - v_i t)$ for each $\pi(t)$

once again via superposition theorem,

 $\pi(t)$ has mean measure of $\lambda_t(x) = \sum_{i=1}^K p_i \lambda(x - v_i t)$

$\therefore$ if cars form a poisson process w/ mean measure λ at time 0, they form a poisson process at some later time t .

$\hookrightarrow$ problem: too restrictive $\rightarrow$ hence mark each points s.t the marks are independent.

2. marking theorem.

idealet π : poisson process on S with mean measure μ .

$\hookrightarrow$ each point x of random set π is associated w/ random variable m_x (marking of x) s.t the distribution depends on x but not π .

 x^* : random point (x, m_x) in $S \times M$ π^* : totality of points $x^* = (x, m_x); x \in \pi \hookrightarrow$ poisson process on $S \times M$

marking theorem

random countable subset π^* is a poisson process on the product space $S \times M$ w/ mean measure given by $M^*(C) = \iint_{(x, mx) \in C} p(x, dm) M(dx)$

proof) let $I^* = \sum_{x \in \pi} f(x, mx)$ s.t.

$$E(e^{-I^*} | \pi) = \prod_{x \in \pi} E(e^{-f(x, mx)} | \pi) = \prod_{x \in \pi} \int_M e^{-f(x, m)} p(x, dm)$$

$$\text{let } f_*(x) = -\log \int_M e^{-f(x, mx)} p(x, dm)$$

recall from Campbell's theorem that

$$E(e^{-I_f}) = \exp(-\int_S (1 - e^{-f(x)}) M(dx))$$

by replacing I_f and $f(x)$ with I^* and $f_*(x)$, we obtain

$$\begin{aligned} E(e^{-I^*}) &= \exp(-\int_S (1 - e^{-f_*(x)}) M(dx)) \\ &= \exp(-\int_S \int_M (1 - e^{-f(x, mx)}) M(dx) p(x, dm)) \\ &= \exp(-\int_{S \times M} (1 - e^{-f}) dM^*) \end{aligned}$$

which shows that π^* is also a poisson process w/ mean measure M^*

corollary

the marks of $x(mx)$ form a poisson process on M .

proof) let $C = S \times B$.

$$\text{since } M^*(C) = \iint_{(x, mx) \in C} p(x, dm) M(dx), \quad \mu(B) = \int_S M(dx) p(x, dm)$$

generalization of colouring theorem

when the marks take on only K different values ($K < \infty$), via marking theorem, we may conclude that the points with i th colour forms a poisson process π_i with mean measure $M_i(A) = \int_A M(dx) p(x, i \text{th})$ which $p(x, i \text{th}) = 1/K$ $\therefore$ probability of colouring now varies with x .

example



large flat pavement ($\mathbb{R}^2$)

more rain $\rightarrow$ overlap $\rightarrow$ irregular wet areas formed

model into poisson process in $\mathbb{R}^2 \times (0, \infty)$

let (x, tx) : drop that falls at point x at time $t \geq x$

then points form a poisson process $\pi(t)$ on $\mathbb{R}^2$

s'pose uniform space & time s.t mean measure has uniform density λ

$\Rightarrow \pi(t)$ has homogeneous poisson process on $\mathbb{R}^2$ w/ density $\lambda(t)$
 suppose r_x is radii of wet circles at time t
 $\Rightarrow (x, t, r_x)$ forms a poisson process in $\mathbb{R}^2 \times (0, \infty) \times (0, \infty)$

3. applications of Campbell's theorem

Campbell's theorem

determines distribution of the sum $\sum f(x)$ over a poisson process of points x .
 want to apply Campbell's theorem in m_x possible since m_x forms a poisson process on M

$$\text{Then, } E(e^{-\sum m_x}) = \exp(-\int_M (1 - e^{-m}) M_m(dm)) \\ = \exp(-\int_M (1 - e^{-m}) M(dx) p(x, dm))$$

example

consider the universe. regard galaxies as point of mass with density $\lambda(x)$
 $\Rightarrow$ mass is independent random variable m_x

the mass m_x of a galaxy at x having density $p(x, m)$ in $m > 0$
 Then (x, m_x) forms poisson process π^* in $\mathbb{R}^3 \times (0, \infty)$ with density $\lambda^*(x, m) = \lambda(x)p(x, m)$
 let $F_j = \sum_{x \in \pi} \frac{G m_x x_j}{(x_1^2 + x_2^2 + x_3^2)^{3/2}} \rightarrow$ gravitational force of (x_1, x_2, x_3) @ origin

via corollary of Campbell's theorem, we have

$E(e^{it_1 F_1 + i t_2 F_2 + i t_3 F_3}) = \exp(\int_{\mathbb{R}^3} \int_0^\infty (e^{i t \cdot \psi(x)} - 1) \lambda(x) p(x, m) dx dm)$ where ψ is a vector function w/ components $\psi_j(x) = (x_1^2 + x_2^2 + x_3^2)^{-3/2} x_j$ and $t \cdot \psi$ is scalar product
 this is valid when $\sum_j = \sum_{x \in \pi} f_j(x)$ converges with probability 1.

we may state that the sum is absolutely convergent and hence may conclude that $\int_{\mathbb{R}^3} m |\psi(x)| \lambda(x) p(x, m) dx dm < \infty$

we can then calculate the means, variances and covariances as:

let $\bar{m}(x) = \int m p(x, m) dm$ denotes the expected mass of galaxy at x .

$$\text{Then, } \int_{\mathbb{R}^3} \frac{\bar{m}(x) \lambda(x)}{x_1^2 + x_2^2 + x_3^2} dx < \infty$$

$$\text{Hence, } E(F_j) = \int \int G m \psi_j(x) \lambda(x) p(x, m) dx dm = G \int_{\mathbb{R}^3} \bar{m}(x) \psi_j(x) \lambda(x) dx$$

$$\text{and } \text{cov}(F_j, F_k) = \int \int G m \psi_j(x) G m \psi_k(x) \lambda(x) p(x, m) dx dm$$

$$= G^2 \int_{\mathbb{R}^3} \bar{m}_2(x) \psi_j(x) \psi_k(x) \lambda(x) dx \text{ where } \bar{m}_2(x) = \int_0^\infty m^2 p(x, m) dm$$

4. motorway 1

on a wide motorway where overtaking is not restricted,
assume: independent movement of cars

① $t=0$: cars are points of poisson process π on $\mathbb{R}$ with rate function $\lambda(x)$

② $t=?$: via marking and mapping theorem, cars form a poisson process $\pi(t)$ at some later time t .

let $Y_t(x)$ as position at some time t . (depends on current position)

via mapping theorem, $Y_t(x)$ form a poisson process $\pi(t)$ on $\mathbb{R}$

via marking theorem, $(x, Y_t(x))$ form a poisson process π^* on $\mathbb{R}^2$

$\therefore$ speed v_i does not have to remain constant & probability of taking speed v_i varies w/ the car's position.

(only holds when vehicle is independent of one another)

Joint distribution of $\pi(t)$ for different t

let $M = \mathbb{R}^n$

By marking theorem, $\{ (x, Y_{t_1}(x), Y_{t_2}(x), \dots, Y_{t_n}(x)) : x \in \pi \}$ is a poisson process on $\mathbb{R}^{n+1}$ for any $t_1, t_2, \dots, t_n$

↳ a.k.a. $\{ (x, Y(x)) : x \in \pi \}$ is a poisson process on $\mathbb{R} \times M$

suppose conditional distribution of $Y_t(x)$ has probability density of $p_t(x, \cdot)$

Then, π^* on $\mathbb{R}^2$ has rate $\lambda(x) p_t(x, y) \rightarrow \lambda_t(y) = \int_{-\infty}^{\infty} \lambda(x) p_t(x, y) dx$

example

suppose cars w/ constant velocities' drawn randomly from a velocity distribution w/ probability density $g(v)$

Then, $p_t(x, y) = t^{-1} g((y-x)/t)$

replacing x with $y-vt$, we obtain $\lambda_t(y) = \int_{-\infty}^{\infty} \lambda(y-vt) g(v) dv$.

* only conserved when cars move independently

5. Displacement theorem

Idea

generalization of previous conclusions for a state space of high dimension.

displacement theorem

let π : poisson process on $\mathbb{R}^d$ w/ rate function $\lambda(x)$

suppose points of π are randomly displaced s.t displacement of different points are independent & distribution of ~~the~~ displaced position of a point of π at $x=x$ has probability density $p(x, \cdot)$

$\Rightarrow$ displaced points form poisson process π' with rate function λ' given by

$$\lambda'(y) = \int_{\mathbb{R}^d} \lambda(x) p(x, y) dx$$

* if $\lambda(x)$ is constant λ and if $p(x, y)$ is a function of $y-x \Rightarrow \lambda'(y) = \lambda \forall y$

proof) supposition $\Rightarrow$ displaced positions Y_x of points x of π defines marking of π .

via marking theorem, $\pi^* = \{Y_x; x \in \pi\}$ is a poisson process on $\mathbb{R}^{2d}$

via mapping theorem, $\pi' = \{Y_x; x \in \pi\}$ is a poisson process on $\mathbb{R}^{2d}$

$\Rightarrow$ mean measure of π^* has density $\lambda(x)p(x, y)$

$\Rightarrow$ mean measure of π' has density $\lambda'(y) = \int_{\mathbb{R}^d} \lambda(x) p(x, y) dx$.

if $\lambda(x) = \lambda$ & $p(x, y) = g(y-x)$,

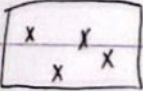
$$\lambda'(y) = \int_{\mathbb{R}^d} \lambda(x) p(x, y) dx = \lambda \int g(y-x) dx = \lambda \int g(x) dx = \lambda \int p(0, x) dx = \lambda$$

application

- poisson process can be an attractive model for random displacements including spatial distribution of animal or plant population.

$\hookrightarrow$ carried out to point of contradiction by Feisenstetn.

contradiction

suppose  area of plane where points = position of plants $\rightarrow$ plants die and distribute seeds around themselves.

if) at least one daughter produced

via displacement theorem, homogeneous poisson process of parent plants produce homogeneous poisson process of daughters.

if) many offsprings produced, $\hookrightarrow$ false statement.

assume N_x daughters born to plant at x has poisson distribution with mean $\phi(x)$ & their positions are $x + p_i$ ($i=1,2,3,\dots,N_x$) where p_i are independent w/ probability $g(x, \cdot)$ on $\mathbb{R}^2$

$\Rightarrow p_i$ form poisson process with rate $\phi(x)g(x, \cdot)$

via mapping theorem, daughters of x form poisson process w/ rate $\lambda(y) = \phi(x)g(x, y-x)$

suppose offsprings of diff plants behave independently.

then, daughters consists of superposition of independent poisson process

via superposition theorem, daughters form poisson process π' w/ rate $\lambda'(y) = \sum \phi(x)g(x, y-x)$
all conditional on position x

if) parents lie at points x of random set π .

$\hookrightarrow$ conditional distribution of π' given π is that of a poisson process w/ rate

$$\lambda'(y) = \sum \phi(x)g(x, y-x)$$

$\hookrightarrow$ unconditional distribution of π' : only in trivial cases be poisson process

$\therefore$ not every generation can be a poisson process

6. Motorway II (orbital motorway)

consider M25 - w/ numerous radial routes developed s.t some are faster than the other and intersects with one another at random collection of points.

q) How many radial routes will remain in use if some slower roads fall into disuse

suppose $\bullet$ radial routes meet the motorway in points of a poisson process where



position described by angular coordinate θ around the circle

$\bullet$ process has uniform rate λ when $0 \leq \theta \leq 2\pi$.

$\bullet$ travel time from centre to point θ is random variable $\tau_\theta \Rightarrow$ the τ_θ for different θ being independent w/ probability density $f(t)$

$\hookrightarrow \tau_\theta$ form marking of π .

$\hookrightarrow$ points (θ, τ_θ) form poisson process π^* on the plane $[0, 2\pi) \times (0, \infty)$ with rate of $\lambda(\theta, \tau) = \lambda f(\tau)$

let w be the angular velocity s.t $2\pi/w$ is the total circumnavigation time

$\hookrightarrow$ route meeting motorway at θ w/ $\tau_\theta = \tau$ is viable if there is no other route (θ', τ') w/ $\tau' + w^{-1}d(\theta, \theta') \leq \tau$ where $d(\theta, \theta') = \min(|\theta - \theta'|, 2\pi - |\theta - \theta'|)$

Hence, probability $V(\theta, \tau)$ that a point $(\theta, \tau) \in \pi^*$ is viable = probability of having no points $(\theta', \tau') \in \pi^*$ in the set C defined by $\tau' + w^{-1}d(\theta, \theta') < \tau$.

$$\Rightarrow V(\theta, \tau) = \exp\left(-\int_C \Lambda(\tau') d\theta' d\tau'\right) \\ = \exp\left(-2\lambda \int_0^\tau \min(w(\tau-t), \pi) f(t) dt\right)$$

We can see this as colouring:

points $(\theta, \tau) \in \pi^*$ coloured with probability $V(\theta, \tau)$

but, since the events aren't independent, we can't assume that the points form a poisson process.

expected number of viable routes

let N be total number of points of π^* w/ distribution $P(2\pi\lambda)$

conditional upon N , (θ_n, τ_n) $n=1, 2, \dots, N$ of π^* are independent w/ probability density $f(\tau)/2\pi$.

$\Rightarrow$ conditional expectation of number of coloured points given N

$$= N \int_0^{2\pi} \int_0^\infty \frac{f(\tau)}{2\pi} V(\theta, \tau) d\theta d\tau$$

$$\text{so that } V = 2\pi\lambda \int_0^{2\pi} \int_0^\infty \frac{f(\tau)}{2\pi} V(\theta, \tau) d\theta d\tau = \int_0^{2\pi} \int_0^\infty \Lambda(\theta, \tau) V(\theta, \tau) d\theta d\tau.$$

then, by substitution, we have

$$V = 2\pi\lambda \int_0^{2\pi} f(\tau) \exp\left(-2\lambda \int_0^\tau \min(w(\tau-t), \pi) f(t) dt\right) d\tau$$